# Chapter 37
# Crypto Derivatives

Chapter 36 worked through markets where the valuation machinery of Parts II and III fails, and in each case it failed because of the underlying: electricity cannot be stored, an emissions allowance has a supply a legislature sets, a degree-day index cannot be held at all.

This chapter's underlying breaks none of those rules. A bitcoin is storable at essentially no cost, perfectly fungible, divisible to eight decimals, pays no dividend, has no convenience yield, no seasonality, and no delivery constraint. By the standards of Chapter 36 it is the best-behaved underlying in the part—better behaved than crude oil, which has to be put somewhere, and far better behaved than an allowance whose validity is a political question.

*So Chapter 5's carry relation applies without modification.* With no income and no storage cost, the arbitrage-free futures price is $F = Se^{rT}$, the cash-and-carry arbitrage can be run in both directions, and the basis trade of Chapter 6 transfers across with the algebra unchanged. When market prices depart from that relation, as Section 37.1 shows they do by several percentage points a year, the reason is not the underlying.

What is missing is everything around the contract. There is no central counterparty, no segregation of client collateral, no daily settlement cycle, because the offshore market never closes and so has no overnight pause in which to collect margin or manage a default, no enforceable netting opinion of the kind Chapter 27 called worth more than any risk model, and no lender of last resort. The machinery of Parts II and III assumes an infrastructure it never names, and this is the market where that assumption is visible because it is absent.

The distinction between the contract and the infrastructure around it organizes the chapter, and it is not a technicality. The largest custody failure in the market's history was not caused by a mispricing, a model failure, or a move in the underlying, and Section 37.10 argues that describing it any other way misplaces the lesson.

Sections 37.1 to 37.5 take the contracts in turn: listed futures and their basis, the effect of a staking yield, perpetual futures, options, and the inverse contract with its choice of numeraire. Section 37.6 looks at the producer and Section 37.7 at the spot ETFs that narrowed the gap to the regulated market; Sections 37.8 and 37.9 are the decentralized venue and the settlement asset. Section 37.10 is the custody failure, and Section 37.11 says who trades these derivatives and why.

## 37.1 LISTED FUTURES AND THE BASIS

Regulated futures on bitcoin have traded since 2017, and the offshore venues list futures on hundreds of tokens. The pricing is Chapter 5's, unchanged.

**Example 37.1 The basis trade, transplanted**

Spot at \$60,000 and the three-month futures price at \$61,800. Crypto markets quote the basis the other way round from Section 3.3's spot minus futures, and as a percentage: the futures premium over spot, $(F - S)/S$. Here that is $1{,}800/60{,}000 = 3.00\%$ over the quarter, or **12.55% annualized** with quarterly compounding $(1.03^4 - 1)$, against a dollar financing cost of 5% a year on the same

compounding basis. In Section 5.9's terms that 5% is the whole cost of carry, since a coin pays no income and costs nothing to store, while the futures contract is priced as if the cost of carry were 12.55%. Buying spot, selling the futures contract and waiting therefore earns the gap: **7.55%** a year over the cost of carrying the position.

Traders rarely put up the full position. Leverage here is the position's notional divided by the capital posted against it (for one unit of a futures contract, the futures price divided by the margin on it), so 20% initial margin is 5× leverage, and the return on that capital is the 7.55% multiplied by the leverage:

| Initial margin | Leverage | Return on capital |
|---|---|---|
| 50% | 2.0× | 15.1% |
| 35% | 2.9× | 21.6% |
| 20% | 5.0× | 37.8% |

This is Section 6.5's Treasury basis trade with the numbers moved up by two orders of magnitude, and the reason the spread is 12.55% rather than a few basis points is not that the arbitrage is harder. It is that three things Chapter 6 took for granted are absent: *there is no CCP*, so the position is an unsecured exposure to a venue; margin is set at the venue's discretion rather than by a published risk model, and can be raised without notice; and *the collateral is held by the counterparty*, so the trade's return is conditional on the counterparty surviving the quarter.

The basis is therefore not an inefficiency but a price. This is Section 5.15's observation that arbitrage requires a balance sheet, in a market where it also requires trusting the venue with the balance sheet.

The return in Example 37.1 does not show the trade's main risk, which is a shortage of cash rather than a loss of value.

### Example 37.2 A hedged trade that a rally can close

The basis trade has no net exposure to the price: the spot leg and the short futures position move together, so the profit is locked in at entry. *The cash flows are not flat.* The futures position is margined daily; the spot is not. So a rally calls cash on one leg against an unrealized gain on the other.

At $60,000 spot the futures price is $61,800, and 20% initial margin on the contract is $12,360 per unit. The cash called is the variation margin the short futures position pays as the futures price rises; with the basis held at 3%, a rally of $x$ raises the futures price by $x \times \$61{,}800$:

| Rally | Cash called | Share of margin | Value of the position |
|---|---|---|---|
| +10% | $6,180 | 50% | unchanged |
| +20% | $12,360 | 100% | unchanged |
| +33% | $20,394 | 165% | unchanged |
| +50% | $30,900 | 250% | unchanged |

A 20% rally, the margin rate itself, exhausts the margin on a position that has not lost a cent. The trade is liquidated for want of cash while the thing it is hedged against is working exactly as intended, and the liquidation forfeits the 7.55%, which was locked in only for a position held to expiration. The futures loss is realized in cash at once, while the offsetting gain on the coin exists only as a mark until the coin is sold; sold alongside, it recovers nearly all of that loss but none of the carry, and kept, it is an unhedged long.

This is Section 8.9's margin call as a liquidity risk, and the reader has met it twice already: it is the problem that brought down Metallgesellschaft's oil hedge, and it is why Chapter 8 insisted that a hedge which is correct in value and wrong in timing is not a hedge. What this market adds is that the venue sets the margin, so the schedule in the table is not a parameter of the trade but a decision someone else may revise mid-position.

Hence the leverage in Example 37.1 is not free. At 5× the headline return is 37.8%, and the

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

rally that ends the trade is 20%. At the 70% volatility this market commonly prices (Example 37.7 below), a coin with no expected drift in its price rises 20% at some point within a quarter with a probability of about 55%. The trade is more likely than not to be liquidated before the futures contract expires and the 7.55% is earned.

The direction of the basis is informative in the same way Chapter 36's curve was. A persistent premium says leveraged long demand exceeds the capital willing to finance it, and the premium collapses, occasionally inverting, when that demand unwinds. It shows how traders are positioned, measured by a market price rather than a survey, and it is easier to read than Chapter 36's convenience yield. That yield is never quoted: it has to be backed out of the curve as the residual left after interest and storage costs. Here the premium itself is quoted, and because a coin pays no income and costs nothing to store, subtracting the dollar interest rate leaves the excess over carry of Example 37.1 with nothing else to estimate.

## 37.2 WHEN THE UNDERLYING PAYS A YIELD

This chapter opened by saying the underlying pays nothing. That is true of bitcoin and false of the second-largest asset in the market, and the correction is worth making because it moves the carry relation from Chapter 5's case of an asset that pays nothing to its case of an asset with a known yield.

A **proof-of-stake** asset pays its holder for committing coin to validating the network. That is a continuous yield on the asset itself, so the forward price is Chapter 5's known-yield case:

$$F = Se^{(r-q)T}. \tag{37.1}$$

The yield $q$ is not set by anyone's discretion. On Ethereum the protocol determines it, and it has two parts. The larger is newly issued coin paid to validators, at a rate that falls as more coin is staked: approximately $166.3/\sqrt{N}$ a year when $N$ ether in total is staked, or about 2.6% at the roughly 41 million staked in 2026. The more coin is committed, the thinner the reward is spread, so the yield regulates how much of the supply is staked. The smaller part is the transaction fees and the payments for ordering transactions that a validator collects when it proposes a block; these vary with network activity and typically add a few tenths of a percent. A staking provider keeps a commission, commonly around 10%, out of both.

Two consequences follow for (37.1). First, $q$ is not known in advance, since both the amount staked and the network's activity move; a forward embeds the yield expected over its life, not a contractual rate. Second, the issuance part is paid in new coin, so it is partly a transfer from holders who do not stake, whose share of the supply shrinks. Unlike a dividend, it is not paid out of anyone's earnings, and a holder who does not stake is diluted rather than merely forgoing income.

**Example 37.3 A dividend the asset did not previously have**

Spot \$3,000, a 5% dollar rate and the staking yield $q$, both continuously compounded, and three months, so the forward is $3{,}000\,e^{(0.05-q)\times 0.25}$:

| Staking yield $q$ | Forward | Quarterly basis | Annualized |
|---|---|---|---|
| 0.0% | \$3,037.74 | +1.258% | +5.13% |
| 2.5% | \$3,018.81 | +0.627% | +2.53% |
| 3.5% | \$3,011.27 | +0.376% | +1.51% |
| 5.0% | \$3,000.00 | 0.000% | 0.00% |
| 7.0% | \$2,985.04 | −0.499% | −1.98% |

The basis is computed as in Example 37.1: the premium $(F - S)/S$ over the quarter, annualized

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

by compounding four quarters. At $q = r$ the futures price equals spot, and above it the curve inverts. The inversion says nothing about scarcity. The asset now pays a yield, and a persistent backwardation that would signal scarcity in Chapter 36 signals here only that the staking yield exceeds the funding rate.

So Example 37.1's basis is not a clean positioning indicator for a staked asset. The observed premium mixes leveraged demand with the yield, and a desk reading it as sentiment will misread it whenever $q$ moves. The two are separable only if the staking yield is quoted independently, which it is, so the correction is available and is routinely not made.

The first US spot ether ETFs of Section 37.7 were launched without staking, and that has a cost. A staked fund followed in March 2026: BlackRock's trust stakes 70 to 95% of its ether and keeps the rest unstaked to meet redemptions. A fund holding coin without staking it forgoes the yield: on a $10 billion fund at 3%, that is $300 million a year, and because most of the yield is new issuance, the fund's holders are diluted by it rather than merely missing it. What the fund buys in exchange is liquidity.

*Staked coin is not immediately deliverable.* Exiting a validating position takes a queue (usually short, but longer when many exit at once, while the entry queue can run to weeks), so the deliverable supply is smaller than the outstanding supply. A fund that must meet redemptions daily cannot hold all its assets behind any queue, which is why a staked fund keeps an unstaked sleeve, and the cash-and-carry arbitrage of Section 37.1 needs unstaked coin to close.

This qualifies the chapter's opening claim in a specific way. The underlying is storable at no cost *and* it now offers a yield for making it temporarily undeliverable, so a holder faces a genuine choice between earning $q$ and retaining delivery capacity. That is Chapter 36's convenience yield reappearing—not as an inference from a curve, but as a posted rate a holder gives up to keep the asset liquid.

## 37.3 PERPETUAL FUTURES

A **perpetual futures** contract (a **perpetual** for short) has no maturity. It never delivers and never expires, so nothing forces convergence to spot. *What replaces convergence is a payment*: at fixed intervals, usually every eight hours (some venues hourly), longs pay shorts a **funding rate** proportional to the gap between the perpetual's price and a spot index, and when the perpetual trades below the index the payment reverses.

The mechanism deserves a moment because it inverts the usual construction. A dated futures contract is held to spot by the *prospect* of delivery, and the cost of carry is embedded in the forward price. A perpetual is held to spot by an explicit cash flow instead, and that cash flow takes the place of the premium in a dated contract. A long in a dated contract pays the carry through the price: it buys at a premium over spot, and the premium shrinks to nothing by delivery. A perpetual's price never converges, so there is no premium to shrink, and without a payment a long perpetual would be leveraged exposure to the coin with nothing paid for the financing. The funding rate is that payment. It is not a cost of carry in Chapter 5's sense, a quantity fixed by interest rates and storage, but a price set by the demand for leverage. It has to cover at least what an arbitrageur who sells the perpetual and holds the coin pays to hold it, and on top of that it carries whatever premium leveraged longs pay, just as the dated basis of Example 37.1 does.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

**Example 37.4 Funding is the basis, quoted differently**

Funding is paid every eight hours, so there are $3 \times 365 = 1{,}095$ periods a year:

| Funding per 8 hours | Annualized | Equivalent 3-month premium |
|---:|---:|---:|
| +0.0100% | +10.95% | +2.74% |
| +0.0300% | +32.85% | +8.21% |
| +0.0500% | +54.75% | +13.69% |
| −0.0200% | −21.90% | −5.47% |

A quoted funding rate of 0.01% per eight hours and a three-month futures contract at a 2.74% premium are the same statement about the price of a levered long, and a desk running both should see them agree. When they do not, the difference is the cost of the two venues' credit and margin terms rather than anything about bitcoin.

In one respect this market reports more than the regulated one does. Chapter 5 had to *infer* a convenience yield as the residual that made an equation hold, and Chapter 36 spent a section extracting one from a curve. The corresponding quantity here, the price of holding a levered long, *is printed every eight hours* rather than backed out of a curve.

The perpetual exists because the infrastructure for delivery does not. A dated contract requires a settlement process, a delivery window, and a mechanism for failures; a contract that never delivers requires none of them. The instrument is an adaptation to missing infrastructure, which is the chapter's theme in a single product.

**Liquidation replaces the margin call.** There is no cure period, and no one at a broker deciding whether to give the client more time. When margin falls below a threshold the venue closes the position at once. In traditional markets the same trigger produces a margin call instead (from the broker on exchange-traded futures, from the counterparty under a credit support annex on OTC derivatives, from the broker on a securities margin account), and a margin call starts a clock: the client has about a day to raise cash, and can see whether the move reverses before committing it. The difference lies with the venue, not the coin, since bitcoin futures on CME carry margin calls like any other futures contract. On an offshore venue the trigger ends the position. The client can add collateral at any moment up to that point, but must do so before knowing whether it will be needed, so protection means collateral posted early and left idle, or a position watched around the clock—and a fast market can cross the threshold before anyone can act. Section 8.9's collateral spiral runs here in minutes rather than days, and because the liquidation levels of leveraged positions cluster at round prices, the forced flow is a known function of the price, which is Section 35.7's barrier clustering in another market, with one difference that Example 37.5 draws out.

**Example 37.5 A cascade with published barriers**

Take \$12 billion of long-biased perpetual open interest, distributed across the leverage the venues actually offer. A position at $\ell$ times leverage has margin worth $1/\ell$ of its notional, so it liquidates on a fall of roughly $1/\ell$ (a little less in practice, since the venue closes it before the margin is entirely gone):

| Leverage | Liquidates on | Notional in tier | Cumulative forced selling |
|---:|---:|---:|---:|
| 25× | −4% | \$1.2bn | \$1.2bn |
| 20× | −5% | \$1.8bn | \$3.0bn |
| 10× | −10% | \$4.2bn | \$7.2bn |
| 5× | −20% | \$3.6bn | \$10.8bn |

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

The table assumes that every position's leverage is measured from the current price and that no one adds margin as the price falls.

A 10% decline mechanically forces $7.2 billion of selling, which is 60% of the open interest, and the selling arrives into a market already falling. *The second-round effect is the point*: that flow pushes the price further, which triggers the next tier. It is the same feedback as the issuer hedging of Section 35.6, except that no issuer decides when to trade: the venue's engine does.

But the disclosure is the opposite of Section 35.7's. Autocallable barriers cluster because issuers independently choose round fractions, and *no issuer can see the aggregate*; that was the whole difficulty. Here open interest and leverage tiers are *published by the venues in real time*, so the liquidation profile is computable by anyone who wants it.

*That does not make the cascade smaller.* It changes who gains from the cascade: the flow is visible in advance to the traders best placed to trade ahead of it, and the resulting anticipation is itself part of why the moves are abrupt. Transparency relocated the problem rather than removing it, which is worth setting beside Chapter 2's argument that disclosure is what makes a clearing system safe.

The largest recorded case is October 10, 2025, when, amid new US tariff announcements aimed at China, trackers counted about $19 billion of liquidations in roughly a day, and at least one large venue closed profitable positions by auto-deleveraging.

## 37.4 OPTIONS

Options trade on the offshore venues and, since 2024, on the regulated ETFs of Section 37.7. The pricing is Section 16.5's. The offshore contracts are European and cash-settled, and the venues price and hedge each one against the futures contract of the same expiration, so the model is Black's, with the futures price in place of spot. That choice matters more here than anywhere in Part III. Black's formula needs only the forward, so it takes in the basis of Section 37.1 and the staking yield of Section 37.2 without anyone having to estimate them, which is Section 16.8's rule of getting the forward right. A value computed from spot and the dollar rate assumes a forward of $Se^{rT}$ and misprices the option by the whole basis, as Example 37.6 shows. Time is counted in calendar days over 365, since the market never closes.

**Example 37.6 Pricing off the forward the market trades**

Spot $60,000, the three-month futures price of $61,800 from Example 37.1, a strike of $60,000, 70% volatility and a 5% dollar rate:

| Forward used | Call | Put |
|---|---|---|
| Futures price, $61,800 (Black) | $9,273 | $7,496 |
| $Se^{rT}$, $60,755 | $8,661 | $7,916 |

Pricing from spot and the dollar rate undervalues the call by $612, or 7%, and overvalues the put by $420, or 5%. The two errors are the same mistake: put–call parity, $c - p = e^{-rT}(F - K)$, ties both prices to the forward, and the second row uses the wrong one. The hedge is in the same contract. The call's delta against the futures contract is $e^{-rT}N(d_1) = 0.595$, so a dealer who sells the call buys 0.595 futures contracts per coin.

The implied volatility surface of these options differs from an equity index's in three ways.

The level is high, and the term structure inverts readily. At-the-money implied volatilities of 50% to 80% are ordinary, and short-dated volatility jumps above long-dated in every sharp move, a shape equity indices show mainly in a crisis (Section 19.6).

*The skew changes sign.* Equity index skew has pointed one way since 1987; *crypto skew reverses with the market*, with calls in demand in a rally and puts in demand in a decline. This is the pattern Section 19.1 found in currencies, whose tilt also changes sign as sentiment moves,

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

rather than the one it found in equity indices, and for a similar structural reason: there is no leverage effect (the tendency of a stock's volatility to rise as its price falls, because a falling share price raises the firm's debt relative to its equity) and no natural buyer of protection, so the skew reflects whichever side is currently buying options.

And the volatility risk premium is large and unstable. Section 19.10's two to four volatility points on equity indices is often ten points or more here, which is why so much of the retail product in this market is a covered call in some form. What that level of volatility costs is worth seeing in currency rather than in percentage points.

**Example 37.7 What a 70% volatility costs**

A thirty-day at-the-money straddle on a \$60,000 spot:

| Volatility | Straddle | Share of spot | Break-even move |
|---|---|---|---|
| 15% | \$2,059 | 3.43% | ±3.43% |
| 30% | \$4,116 | 6.86% | ±6.86% |
| 50% | \$6,857 | 11.43% | ±11.43% |
| 70% | \$9,591 | **15.99%** | ±15.99% |
| 90% | \$12,318 | 20.53% | ±20.53% |

An equity index at 15% costs 3.43% of spot to straddle for a month; this market at 70% costs 15.99%. The buyer needs a 16% move in thirty days merely to return the premium, and the seller collects 16% of spot for a month of risk, which is why the yield products of Section 37.11 can advertise returns that look impossible and are not.

*The premium is real and it is not a riskless profit.* To first order, selling variance earns $(\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2)/2$ a year:

| Implied | Realized | Variance premium |
|---|---|---|
| 70% | 60% | +6.5% a year |
| 70% | 55% | +9.4% a year |
| 70% | 75% | −3.6% a year |

Ten volatility points of edge is worth 6.5% a year and five points of adverse surprise costs 3.6%, so the strategy's sign depends on a quantity no one observes in advance, which is Section 22.4's point about variance swaps, in a market where the numbers are several times larger and the estimates far less stable.

## 37.5 INVERSE CONTRACTS AND THE CHOICE OF NUMERAIRE

A large part of this market trades in contracts that settle in the coin rather than in dollars, and the convention is not cosmetic. An **inverse** contract has a notional fixed in dollars, margin posted in the coin, and settlement paid in the coin: a long position of notional $Q$ entered at $F_0$ pays

$$\Pi_{\text{coin}} = Q\left(\frac{1}{F_0} - \frac{1}{S_T}\right). \tag{37.2}$$

This is Section 28.6's numeraire question posed by a contract specification rather than by a model. Convert (37.2) at the terminal price and the dollar payoff is $Q(S_T/F_0 - 1)$—*exactly linear*. Leave it in coin and it is not.

**Example 37.8 Linear in one currency, curved in the other**

A long inverse position, \$100,000 notional, entered at \$60,000:

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

| Price | Move | P&L in coin | P&L in dollars |
|---|---|---|---|
| \$30,000 | −50% | −1.6667 | −\$50,000 |
| \$42,000 | −30% | −0.7143 | −\$30,000 |
| \$48,000 | −20% | −0.4167 | −\$20,000 |
| \$72,000 | +20% | +0.2778 | +\$20,000 |
| \$90,000 | +50% | +0.5556 | +\$50,000 |
| \$120,000 | +100% | +0.8333 | +\$100,000 |

The dollar column is a straight line and the coin column is not. A 50% fall costs **1.6667 coin** while a 50% rise earns only 0.5556, and the coin gain is capped at $Q/F_0 = 1.6667$ however far the price rises while the coin loss is unbounded. A long inverse position is short gamma measured in coin, with $\partial^2 \Pi_{\text{coin}}/\partial S^2 = -2Q/S^3$.

The practical consequence is that margin is computed in the unit where the position is curved. Start with 0.5 coin of margin:

| Price | Equity (coin) | Equity (dollars) |
|---|---|---|
| \$54,000 | +0.3148 | \$17,000 |
| \$48,000 | +0.0833 | \$4,000 |
| \$42,000 | −0.2143 | −\$9,000 |

Equity in coin falls faster than the price does, because the loss is denominated in an asset that is itself losing value, so liquidation arrives earlier than a dollar-margined trader would compute from the same leverage. The collateral and the exposure are the same asset, which is Chapter 27's wrong-way risk in its purest form, written into the contract rather than arising from a correlation.

The convention has a practical origin. An inverse contract needs no dollar leg, so a venue can run it without touching the banking system: the whole contract lives in the asset it references. That mattered when venues could not get bank accounts, and it is why the oldest and deepest contracts in this market are inverse ones while the newer dollar-margined contracts dominate flow from institutions that have banks.

And the pricing follows Chapter 28 mechanically once the numeraire is named. Valuing (37.2) in dollars means valuing a coin-denominated payoff, so the coin is the numeraire, and the change of measure that goes with it is the one Section 30.6 performs for a quanto. The convexity correction that follows is the same object as a quanto adjustment, and a desk that prices inverse contracts by discounting the dollar payoff is dropping it.

## 37.6 THE PRODUCER: MINING AS A SPREAD POSITION

This market has a producer, as Chapter 36's power market does. A miner spends electricity and capital to receive coins, which makes it *long the coin and short power*, which is Section 36.6's spark spread with a different output. The comparison holds closely enough to be useful, and the hedging problem lies where it fails.

**Example 37.9 A miner's break-even, and the hedge the protocol supplies**

A miner produces 500 coins a year against \$14 million of electricity and \$4 million of other costs. *Break-even is \$36,000 a coin*, and because the costs are fixed while revenue moves with the price, the margin moves much more than the price does:

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

| Coin price | Revenue | Costs | Margin | Margin/revenue |
|---|---|---|---|---|
| $30,000 | $15.0m | $18m | −$3.0m | −20.0% |
| $45,000 | $22.5m | $18m | +$4.5m | +20.0% |
| $60,000 | $30.0m | $18m | +$12.0m | +40.0% |
| $90,000 | $45.0m | $18m | +$27.0m | +60.0% |

The output is not a fixed quantity, and that is what separates this from the spark spread. The network issues new coins at a fixed rate, a batch with each block of transactions, and a miner wins blocks in proportion to its share of the **hash rate**, the computing power all miners together apply to the puzzle each block requires. The protocol sets how hard that puzzle is, and its **difficulty adjustment** resets it at intervals so that blocks keep arriving on schedule however much hardware is running. If a price fall drives competitors off, the difficulty falls and the survivor's share rises.

Suppose the price falls from $60,000 to $42,000 and 20% of the network's hash rate leaves:

| | Revenue | Change |
|---|---|---|
| Before | $30.0m | — |
| After, at constant share | $21.0m | −30% |
| After the difficulty adjustment | $26.3m | **−12.5**% |

The protocol converts a 30% revenue shock into a 12.5% one. The miner is long the price, and it is also long the exit of competitors, which happens when the price falls; the two offset, so the exposure is to price × share, and share moves against price.

Three qualifications keep this from being a free hedge:

- the difficulty adjustment comes at intervals of roughly two weeks, so the offset arrives after the cash-flow problem it would have solved;
- the protection comes from the miners who leave, and those are the high-cost ones, so a high-cost miner is near the exit itself when it needs protection most, while the low-cost miner, which needs it least, gains the most; and
- the hedge works through quantity, so a forward sale of a fixed amount fits neither direction: after a fall the miner produces more than it sold and the surplus is unhedged, and after a rise it produces less and must buy coin at the high price to deliver.

*This explains how miners hedge.* They sell a fraction of expected output forward rather than all of it, and they buy puts more often than they sell forwards, because a forward that must be delivered into is a short position in a market where the quantity is itself uncertain. *This is volume risk.* The contract built for it is the swing contract of Section 36.5, which lets its holder choose how much to take or deliver within limits; a miner would want the delivering side of one, and no such contract trades on coin output. So miners approximate it with a strip of listed puts and accept the basis.

Power contracting is the other half of the position and it is where the interesting structures are. A miner with an interruptible power contract has sold an option to the grid: it agrees to shut down on demand in exchange for a lower price. Interruptible contracts are old (aluminum smelters have signed them for decades), but a miner is an unusually good counterparty for one, because what it gives up by shutting down is coin, which can be valued at any moment: the shutdown cost is observable at every instant as the coin price times the foregone production, so the option's exercise value is marked to a screen. Several grids now contract with miners this way.

## 37.7 SPOT ETFS AND THE NARROWING OF THE GAP

The most consequential development in this market was not a derivative. United States spot bitcoin exchange-traded funds began trading in January 2024, and options on those funds followed in November 2024.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

What that changed is precisely the thing this chapter says is missing. An ETF share sits in ordinary custody, its options clear through the Options Clearing Corporation, the positions net under an enforceable framework, and the collateral is segregated. *None of it altered bitcoin*, which behaved identically before and after. It altered the infrastructure wrapped around it.

The gap between the two venues is therefore measurable, and it is the price of that infrastructure. An option on the ETF and an option on the offshore venue reference nearly the same underlying and differ in counterparty, segregation, netting, and margin treatment. The spread between their implied volatilities is not a modeling puzzle: it is Chapter 27's capital and Chapter 8's collateral, expressed in volatility points, and it is the cleanest available estimate of what the missing infrastructure was worth.

## 37.8 AUTOMATED MARKET MAKERS

A decentralized exchange has no order book. It holds an inventory of two assets and quotes by a rule: the **constant-product** rule keeps $x\,y = L^2$, where $x$ is the inventory of a token and $y$ the inventory of a dollar stablecoin (a token its issuer promises to redeem for one dollar, backed by a reserve of dollar assets, and the settlement asset of Section 37.9), counted here as dollars, and anyone may trade against the pool at the price that rule implies. That price is the ratio of the inventories, $p = y/x$, because along the curve $xy = L^2$ a small trade exchanges $dx$ tokens for $dy = -(y/x)\,dx$ dollars. The constant is written as a square so that $L = \sqrt{xy}$, the geometric mean of the two inventories, measures the pool's size; it is called the pool's **liquidity**, and a deeper pool moves its price less for a trade of a given size. Neither inventory is fixed; only their product is. A buyer of the token pays stablecoin in and takes tokens out, so $y$ rises, $x$ falls and the price $y/x$ goes up, while $L$ changes only when providers deposit or withdraw.

Providers of that inventory are told they are earning fees. They are, and Example 37.10 shows what they are paying for them.

**Example 37.10 Providing liquidity is selling gamma**

At a price $p$ the two conditions $xy = L^2$ and $y/x = p$ pin the inventory to $x = L/\sqrt{p}$ and $y = L\sqrt{p}$, so the position is worth

$$V(p) \;=\; x\,p + y \;=\; 2L\sqrt{p}. \tag{37.3}$$

*Two derivatives tell the whole story.* The delta is $V'(p) = L/\sqrt{p} = x$, *exactly the token inventory*, so the pool is automatically delta-hedged to its own holdings. And

$$V''(p) \;=\; -\frac{L}{2\,p^{3/2}} \;<\; 0,$$

so the position is concave in the price everywhere. *A liquidity provider is short gamma*, and Chapter 18 established what a short gamma position earns and what it loses.

Against simply holding the initial basket, the shortfall (the market calls it *impermanent loss*, a name suggesting the loss will reverse, which it does only if the price returns to where it started) depends only on the price ratio $k$:

$$\mathrm{IL}(k) \;=\; \frac{2\sqrt{k}}{1+k} - 1$$

| **Price ratio** | 0.25× | 0.5× | 0.8× | 1.25× | 2× | 4× |
|---|---|---|---|---|---|---|
| **Shortfall** | −20.00% | −5.72% | −0.62% | −0.62% | −5.72% | −20.00% |

It is symmetric in $k$ and $1/k$, which is the signature of a short straddle: a payoff concave in the underlying is a short position in options.

The cost of being short gamma, the rate at which the pool falls behind the basket and which the market calls its bleed, has a simple closed form. Chapter 18's gamma cost per unit time is

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

$\frac{1}{2}|V''|\sigma^2 p^2 = L\sigma^2\sqrt{p}/4$, and dividing by the position's value $2L\sqrt{p}$:

$$\text{cost per year} \;=\; \frac{\sigma^2}{8} \qquad \text{of the position, whatever the price and whatever the pool size.} \quad (37.4)$$

| **Volatility** | 40% | 60% | 80% | 100% | 120% |
|---|---|---|---|---|---|
| **Fee yield needed to break even** | 2.00% | 4.50% | **8.00%** | 12.50% | 18.00% |

The exact expected shortfall also has a closed form, $1 - e^{-\sigma^2 T/8}$, and it confirms (37.4) and shows where it stops applying: at 80% volatility over three months the exact loss is −1.980% against a predicted −2.000%, and over a year −7.688% against −8.000%. *The bleed is the small-$\sigma^2 T$ expansion of that exponential*, so it always overstates the loss: by 4% of the exact figure at the one-year horizon, but at 120% volatility over three years, where $\sigma^2 T = 4.32$, its 54.0% overstates the exact 41.7% by 29%.

So the question a liquidity provider should ask is the one an option seller asks: is the premium above or below the volatility I am short? At 80% volatility the pool must return 8% a year in fees before the provider is even, and a pool advertising a 6% fee yield on a pair that volatile is advertising a loss.

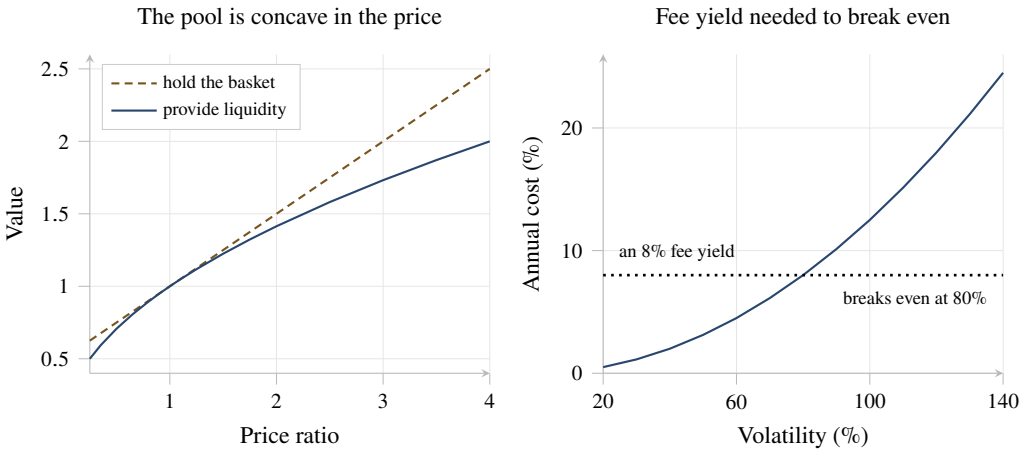



**Figure 37.1:** Left: pool value against holding the same initial basket, normalized to 1. Right: (37.4).

Figure 37.1 is the reason this section is in a derivatives textbook rather than a technology one. The left panel is a written straddle drawn in unfamiliar coordinates, and the concavity is the product, not a defect of it. A provider who understood the position as a short option would ask about the premium; one who understands it as "earning yield" will not.

The concentrated-liquidity variant makes the option clearer. Providers may confine their inventory to a price range, which concentrates the gamma and raises both the fee income and the bleed—and outside the range the position is entirely one asset and the delta is constant. *That is a short straddle with knock-outs*, which is Chapter 34's territory, and it is sold to retail users with no options vocabulary at all.

## 37.9 THE SETTLEMENT ASSET

Almost every trade on the offshore venues settles in a **stablecoin** rather than in dollars, and that changes what the quoted prices mean. A perpetual quoted at 60,000 is quoted in a token that its issuer promises to redeem at a dollar, so the futures price carries the issuer's credit whether or

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

not anyone prices it that way.

The peg is not a law of the market; it is an arbitrage with a cost. If the token trades below a dollar, someone buys it and redeems it at par for the profit. That trade works only while redemption works, which requires the reserve to be liquid, the issuer to be operating, and the redemption window to be open. Remove any of the three and the token stops being a dollar and becomes a claim on a portfolio, valued as Part V values any other claim.

**Example 37.11 What the discount was pricing**

In March 2023 a large stablecoin disclosed that $3.3 billion of its roughly $40 billion reserve, about 8%, sat at a bank that had just failed. Take the rest of the reserve as sound and price the token as a claim:

| Recovery on the exposed reserve | Fair value | Discount |
|---|---|---|
| 100% | 1.0000 | 0.00% |
| 50% | 0.9600 | 4.00% |
| 0% | 0.9200 | 8.00% |

The token traded to about 0.8774, a 12.3% discount, against an arithmetic worst case of 8.00%. *No recovery assumption produces that price*, so the extra 4.3 points were not pricing the disclosed exposure at all.

They were pricing the redemption mechanism itself. The arbitrage that holds the peg needs an open window; a holder who fears the window closes before their turn will sell at any price above what they expect a queue to yield. So the discount decomposes into a credit loss on a known exposure and an option-like charge for access to the redemption facility, and only the first is in the disclosure.

*The link to the rest of the book is direct.* The failure of Silicon Valley Bank (Section 40.14) reached the derivatives market of this chapter through the settlement asset—not through any position, model, or counterparty in it. A trader perfectly hedged in coin terms lost 12.3% on the units the hedge was denominated in, which is Section 7.8's currency basis argument with the funding currency issued by a private firm.

Two consequences follow for anyone marking a book in this market. First, a coin-denominated profit and loss is not a dollar profit and loss, and the two diverge exactly when the market is stressed—which is when the distinction matters. Second, margin posted in a stablecoin is collateral with credit risk and correlation to the very events that trigger margin calls, so Chapter 27's wrong-way risk applies to the collateral rather than to the exposure. A haircut on the settlement asset is the obvious remedy and is almost never applied, because applying it would concede that the unit of account is not a dollar. Regulation is now supplying part of what the haircut did not. The US GENIUS Act of 2025 requires payment-stablecoin reserves of high-quality liquid assets and redemption at par, and in February 2026 the CFTC allowed futures commission merchants to accept stablecoins issued by national trust banks as margin. Whether a clearing member's margin schedule applies a haircut to them is the question to ask.

## 37.10 THE VENUE IS THE COUNTERPARTY

The failure of FTX in November 2022 is usually told as a story about crypto. It is more useful as a story about Chapter 2, because every layer that chapter described was missing and the absences can be listed one by one.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

**Case Study 37.1 A default waterfall with no layers**

Chapter 2 set out the sequence in which a clearing house absorbs a member default. Set that sequence beside what a large crypto venue had in place of each layer:

| Chapter 2's waterfall | The venue |
|---|---|
| Defaulter's initial margin | Held by the venue, unsegregated |
| Defaulter's default fund contribution | None |
| The CCP's own capital, ahead of members | No distinction between the venue and its operator |
| Surviving members' mutualized fund | An "insurance fund" of undisclosed size and composition |
| Assessments and recovery tools | Losses socialized by unilateral decision |

*Bitcoin settled every block throughout.* No contract mispriced, no model failed, and the underlying did what it always does. *Client assets were not segregated*, so a customer's exposure was never to the market: it was to the venue, and the venue was the same legal person as the trading operation on the other side.

Section 27.2 called a jurisdiction's netting opinion (a lawyer's opinion that close-out netting will be enforced if a counterparty fails) worth more to a dealer than any risk model. This market shows what that means by its absence. There was no netting opinion, no bankruptcy-remote structure, and no supervisor to ask for either, so the entire apparatus of Part V (exposure at default, close-out netting, the margin period of risk) had nothing to attach to.

The lesson is not that the instruments were exotic. They were the plainest instruments in this book: futures, perpetuals, and a spot balance. What was exotic was the absence of the infrastructure every earlier chapter assumed.

Where a clearing house would draw on its default fund, a crypto venue uses **auto-deleveraging**, which has no analog in Part I. When a liquidation cannot be filled at a price that leaves the venue whole, the shortfall is met by closing profitable positions on the other side, ranked by their profit and leverage. A trader who is right about the market and fully margined can have the position taken away to fund somebody else's loss, which is Chapter 2's assessment powers without the mutualization, the disclosure, or the governance.

## 37.11 WHO TRADES CRYPTO DERIVATIVES, AND WHY

**Leveraged directional traders** are the dominant flow and the reason perpetuals exist: a contract with no expiration, no delivery, and high leverage is the cheapest way to express a view. Their liquidation levels are the forced flow of Section 37.3, which makes their aggregate position a market-structure variable rather than a private one.

**Basis traders and market-neutral funds** harvest the basis of Example 37.1, and they are the capital that keeps the futures anchored to spot. *They are also the first to leave*, since their return is a spread and their risk is the venue, so the basis widens exactly when the venue is in question.

**Miners** are the natural commercial hedgers, with a revenue stream in coin and costs in fiat and electricity, *the configuration of Chapter 36's generator*, and they hedge it by selling forward and buying puts.

**Companies and funds holding coin** sell covered calls for yield, giving up the price rise above the strike for a premium. It is the same trade, with the same asymmetry, as the yield products of Chapter 35.

**Liquidity providers** supply the inventory of Section 37.8, mostly without pricing the option

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

they have written.

The market's participant mix is unusual in one respect. In every other chapter of this book the natural hedger came first and the speculator followed. Here the speculative flow built the market and the commercial hedgers arrived afterward, which is why the instruments are optimized for leverage rather than for delivery, and why the infrastructure lagged the volume by a decade.

## SUMMARY

The underlying is the easy part and the infrastructure is the hard part. A bitcoin is storable at no cost, fungible, divisible, and pays nothing, so Chapter 5's $F = Se^{rT}$ is the arbitrage-free price. What is absent is the CCP, the segregation, the daily settlement cycle, the netting opinion, and the lender of last resort: the infrastructure Parts II and III assume without naming it.

That explains the prices. A futures basis far wider than the cost of financing is the venue's credit being priced, not an inefficiency. On a staked asset the basis also mixes in the yield, so it no longer reads as positioning alone, and options are priced off the forward the market trades rather than off spot. The perpetual is the only genuinely new contract: with nothing forcing convergence, a funding rate paid every eight hours does delivery's work and prints the price of a levered long that Chapter 5 had to infer. Liquidation replaces the margin call, and because leverage tiers are published the forced flow is computable in advance, which relocates Section 35.7's barrier clustering rather than removing it.

Two conventions carry more risk than they appear to. An inverse contract is linear in dollars and curved in the coin, so liquidation arrives earlier than dollar leverage suggests. And almost every offshore trade settles in a stablecoin, so the quoted price carries an issuer's credit and the value of access to redemption. In both cases collateral and exposure are the same risk: Chapter 27's wrong-way risk sitting in the contract and in the unit of account.

The producer and the liquidity provider carry options they do not price as such. A miner holds Section 36.6's spark spread, long the coin and short power; the difficulty adjustment is an automatic hedge that arrives late and works through quantity, which is why miners buy puts rather than sell forward, and a miner on an interruptible power contract has sold the grid an option to shut it down. Providing liquidity to a constant-product pool is a written straddle that bleeds $\sigma^2/8$ a year whatever the price, so only substantial fees let the provider break even at this market's volatilities.

FTX is a story about Chapter 2. Every waterfall layer was missing, yet bitcoin settled every block throughout: no contract mispriced and no model failed. Client assets were not segregated, so the exposure was to the venue rather than to the market. The regulated route of the 2024 ETFs supplies the missing infrastructure. The instruments were never the problem.

## FURTHER READING

The primary sources in this market divide sharply: venue documentation of funding-rate and liquidation mechanics is precise and worth reading directly, while almost everything written about valuation restates Chapters 5 and 15 with new vocabulary. On the perpetual contract, the mechanism papers are short and the specification documents are shorter, and both are the place to start, before the secondary literature. On automated market makers, Milionis, Moallemi, Roughgarden and Zhang on loss-versus-rebalancing derive the $\sigma^2/8$ rate of Section 37.8 from the Black–Scholes setting, and the connection to Chapter 18's gamma cost is immediate. The FTX bankruptcy filings are the account of what Section 37.10 describes, and they enumerate what was and was not held. On the regulated products, the fund prospectuses set out the custody and creation mechanics that Section 37.7 argues are the whole of the change.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

## PRACTICE QUESTIONS

**37.1** Using Example 37.1, state the annualized basis and the return on capital at 20% initial margin, and give three reasons the spread is wider than Chapter 6's Treasury basis.

**37.2** Using Example 37.2, state the rally that exhausts the posted margin on a fully hedged basis trade, and explain why a position that has lost nothing can be closed.

**37.3** Explain why the leverage in Example 37.1 and the rally in Example 37.2 must be quoted together, and which earlier chapter's case has the same shape.

**37.4** Using Example 37.3, state the forward price and basis at a 3.5% staking yield, and the yield at which the future trades at spot.

**37.5** Explain why a persistent backwardation in a staked asset does not mean what the same shape means in Chapter 36, and what must be quoted to separate the two.

**37.6** Explain what holds a perpetual futures contract to spot in the absence of delivery, and why the mechanism is an adaptation to missing infrastructure.

**37.7** Using Example 37.4, convert a funding rate of 0.03% per eight hours into an annualized rate and into the equivalent three-month futures premium.

**37.8** Explain why liquidation on a crypto venue is not a margin call, and what the client gives up when it replaces one.

**37.9** Using Example 37.5, state the forced selling triggered by a 10% decline, and explain why publishing the liquidation profile does not make the cascade smaller.

**37.10** Explain why crypto skew changes sign with the market while equity index skew does not, and name the chapter whose pattern it resembles.

**37.11** Using Example 37.8, state the coin profit and loss of a long inverse position at a 50% fall and at a 50% rise, and explain why the two are not symmetric.

**37.12** Explain why an inverse contract's valuation is a change of numeraire, and which adjustment a desk drops by discounting the dollar payoff instead.

**37.13** Using Example 37.9, state the break-even coin price, and the revenue change from a 30% price fall before and after the difficulty adjustment.

**37.14** Explain why the difficulty adjustment is only a partial hedge, giving the three qualifications, and why miners prefer puts to forwards.

**37.15** Using Example 37.11, compute the fair value at 50% recovery on the exposed reserve, and explain what the observed 12.3% discount priced that the disclosure did not.

**37.16** Explain why margin posted in a stablecoin is wrong-way risk in the collateral, and what remedy Section 37.9 says is almost never applied.

**37.17** Using Example 37.10, state the shortfall at a price ratio of 2 and of 4, and explain why the function is symmetric in $k$ and $1/k$.

**37.18** Derive the $\sigma^2/8$ bleed of (37.4) from Chapter 18's gamma cost, and explain why it does not depend on the pool size.

**37.19** Explain why the FTX failure is described as a custody failure rather than a market failure, and what the underlying did throughout.

**37.20** Explain what auto-deleveraging is, why it has no analog in Part I, and why a correctly positioned and fully margined trader can still lose the position.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!

## FURTHER QUESTIONS

**37.21** Example 37.4 treats the funding rate and the futures basis as two quotations of one price. Construct the arbitrage that enforces the relation, determine what frictions would let the two persist at different levels, and estimate how large a gap those frictions could sustain.

**37.22** Example 37.5 treats the liquidation profile as static. Model it dynamically, so that the selling from one tier moves the price into the next, determine the condition on the leverage distribution under which the cascade is self-limiting rather than self-sustaining, and state what a venue could change to enforce that condition.

**37.23** Example 37.9 shows that the difficulty adjustment offsets part of a price fall with a lag. Formulate the miner's revenue as a product of price and share, model the share as a delayed response to price, and determine the forward-sale fraction that minimizes the variance of revenue.

**37.24** Example 37.11 decomposes a discount into a credit loss and a charge for redemption access. Build a model of the second component as an option on the window's remaining capacity, calibrate it to the observed 12.3%, and state what the calibrated parameter would have to mean.

**37.25** Example 37.10 values a pool under geometric Brownian motion. Redo the analysis under a jump-diffusion, determine whether the $\sigma^2/8$ result survives, and state what a provider should charge for the jump component that the diffusion formula omits.

**37.26** Concentrated liquidity confines the inventory to a price range. Formulate the resulting position as a combination of Chapter 34's barrier options, derive the fee yield needed to break even as a function of the range width, and determine the width that maximizes expected profit at a given volatility.

**37.27** Section 37.7 argues the ETF-to-offshore volatility spread prices market infrastructure. Design the measurement: state which contracts to compare, what must be controlled for, and how you would separate the infrastructure premium from a pure liquidity difference.

**37.28** The snapshot in Section 37.10 maps a crypto venue onto Chapter 2's waterfall. Design a waterfall that would work for a venue with no members in Chapter 2's sense, state who funds each layer, and identify which layer is hardest to fund and why.

Wulin Suo: WSuo@Queensu.ca. For Personal Use, Do Not Distribute!